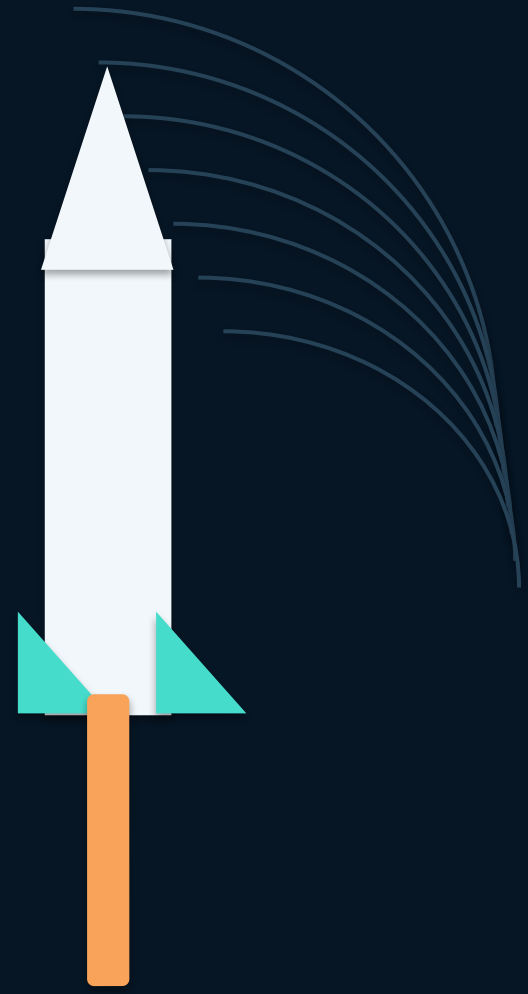
IBM DATA SCIENCE
CAPSTONE PROJECT

SpaceX Falcon 9 First-Stage Landing Prediction

Methods · exploratory analysis · SQL · geospatial insights · machine learning



FINAL REPORT / 25 OCT 2026

HISTORICAL SPACE X LAUNCH DATA

Executive summary

90

Modelling observations

60

Successful landings

66.7% in the 90-flight dataset

83.3%

Held-out test accuracy

15 correct predictions / 18

What we learned

Landing success varies by launch site, payload range and booster generation in the historical data. KSC LC-39A leads the 56-launch dashboard subset.

Decision implication

Four classification algorithms tie at 83.3% test accuracy. Interpretation must include false positives, small sample size and dataset limitations.

Not a current SpaceX performance assessment: source datasets are course-provided historical snapshots.

Sources: Data Wrangling; Dashboard; Machine Learning notebooks.

Research question and outcome definition

Can historical mission and booster characteristics predict whether the Falcon 9 first stage lands successfully?

Target label = 1

Successful first-stage landing / accepted success categories under the IBM data-wrangling lab.

Target label = 0

Failed landing, no landing attempt, or other outcome categorized as unsuccessful in the exercise.

Important distinction: mission success (payload delivered) and first-stage landing success are different outcomes.

Sources: Data Wrangling and Machine Learning notebooks.

Study architecture: four different data views

121

Scraping

Web-derived historical launches · 11 columns

101

SQL

Historical launch-table records for SQL tasks

90

ML / EDA

Labelled observations · 18 test cases

56

Dash / Folium

Launch-site geographic/dashboard subset

> These cohorts have different coverage and naming conventions; their percentages must not be pooled.

Sources: Web Scraping, SQL, Data Wrangling, Folium and Dash notebooks.

Acquisition: API workflow and web scraping

API collection (not verified)

Notebook documents the SpaceX REST API workflow (rockets, launchpads, payloads, cores). The uploaded API notebook has no saved executed outputs.

Web scraping (verified)

BeautifulSoup parses the archived Falcon 9/Falcon Heavy launch tables and saves an 11-column CSV. The notebook reports 121 extracted records.

121

Extracted web records

11

Scraped output columns

—

Executed API results

Sources: API notebook (unexecuted); Web Scraping notebook (executed).

Data wrangling: define the learning target



Cleaning and labelling

Missing-value review, landing-outcome mapping, target Class∈{0,1}, and export of dataset_part_2.csv.

Observed missingness

LandingPad is missing in 28.89% of records; no missing values are reported for the other listed input variables.

The 66.7% figure is the labelled first-stage landing rate in the 90-record modelling cohort.

Source: Data Wrangling(1).ipynb, cells 19–50.

Exploratory analysis with SQL

101

Launch records

A separate IBM SQL exercise table.

45,596 kg

NASA (CRS) payload mass

Total recorded payload mass for this exact customer label.

2,928.4 kg

Falcon 9 v1.1

Average recorded payload mass in SQL dataset.

2015-12-22

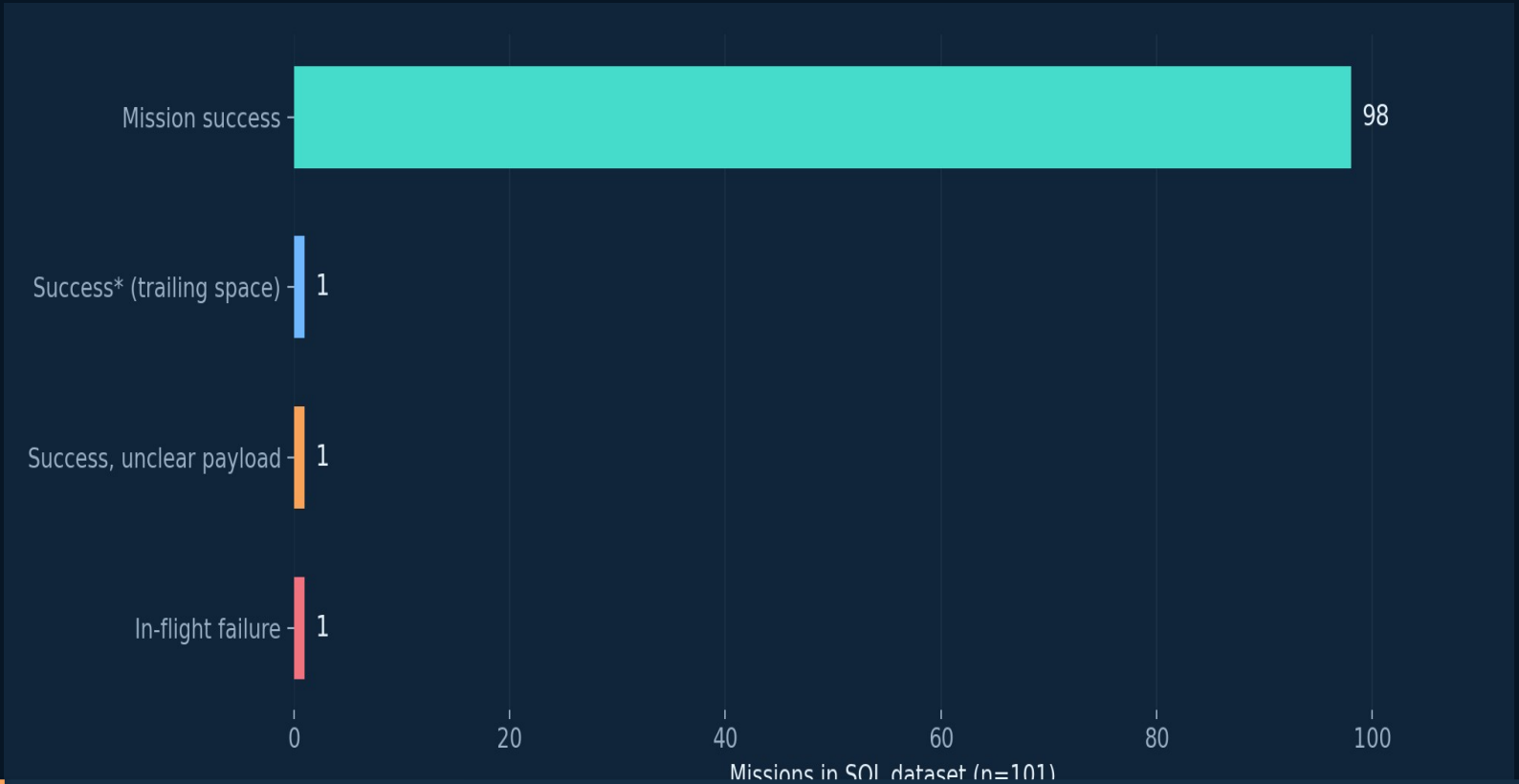
First ground-pad landing

Earliest date of “Success (ground pad)” recorded.

These are values returned by the uploaded notebook queries; the original SQL cohort is not the 90-flight ML cohort.

Source: Complete the EDA with SQL.ipynb, executed query outputs.

Mission outcome is not landing outcome



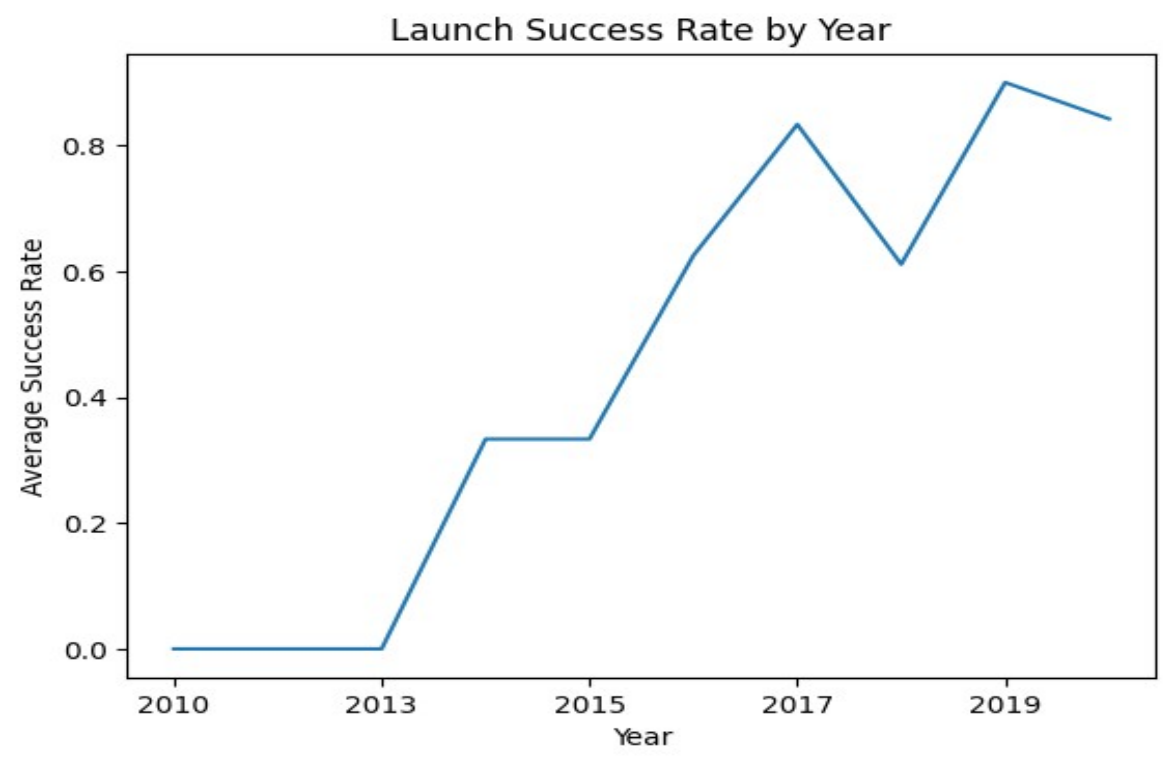
Data quality insight

The SQL results include “Success” and a separate “Success ” value with a trailing space. Category normalization matters before aggregation.

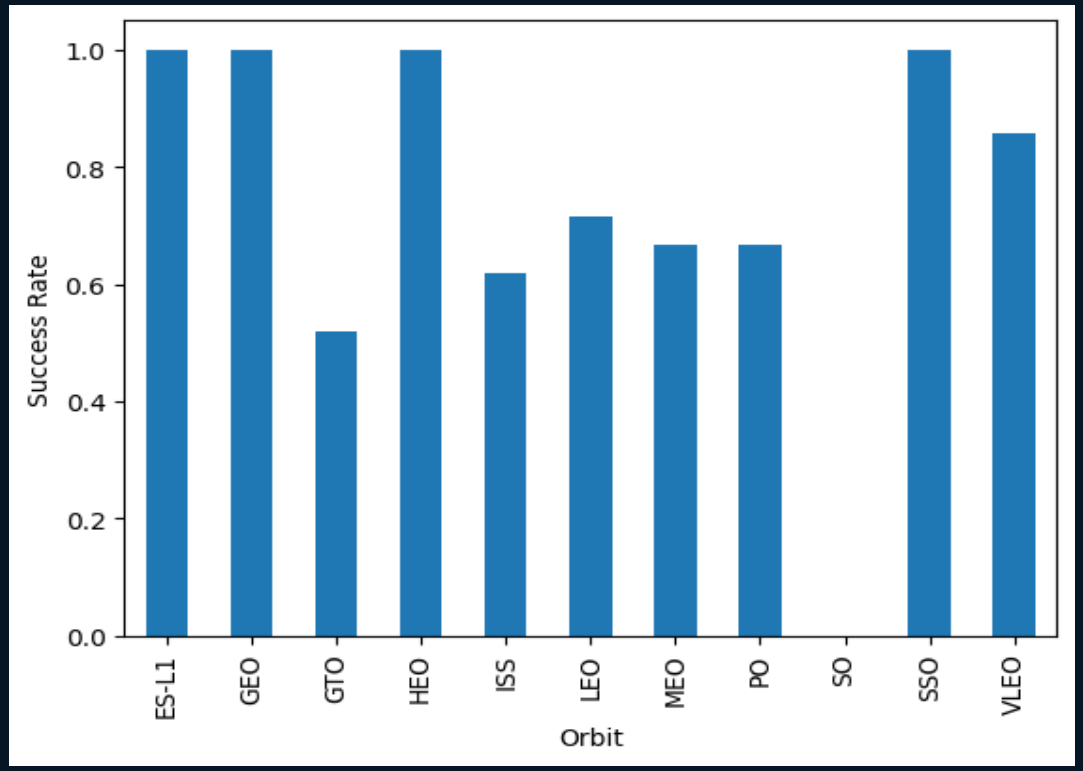
98 exact “Success” records + 1 with trailing whitespace; 1 unclear payload; 1 in-flight failure.

Source: SQL notebook, query grouping Mission_Outcome (101 records).

Exploratory visual analysis: key patterns



Yearly first-stage landing rate in the historic sample

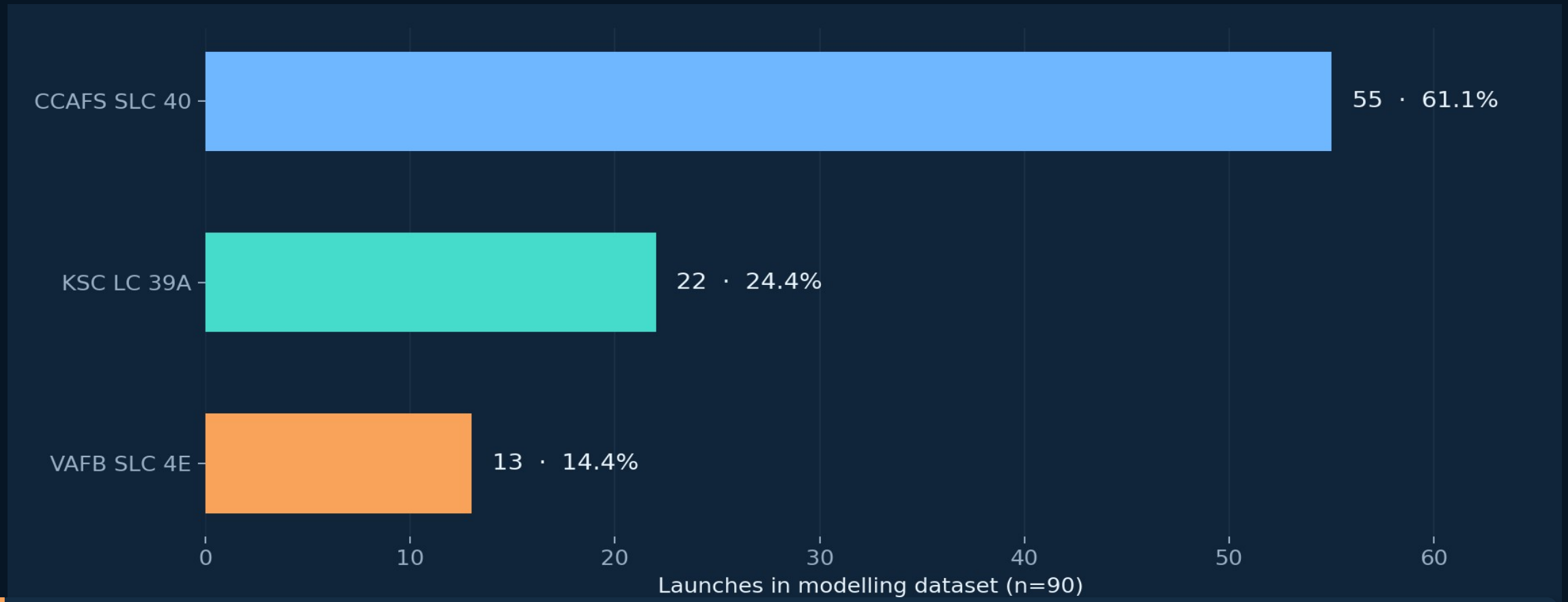


Landing rate varies across orbit categories

> Descriptive associations only: time trends, orbit and payload patterns do not establish causal effects.

Source: EDA with Visualization Lab, executed plot outputs.

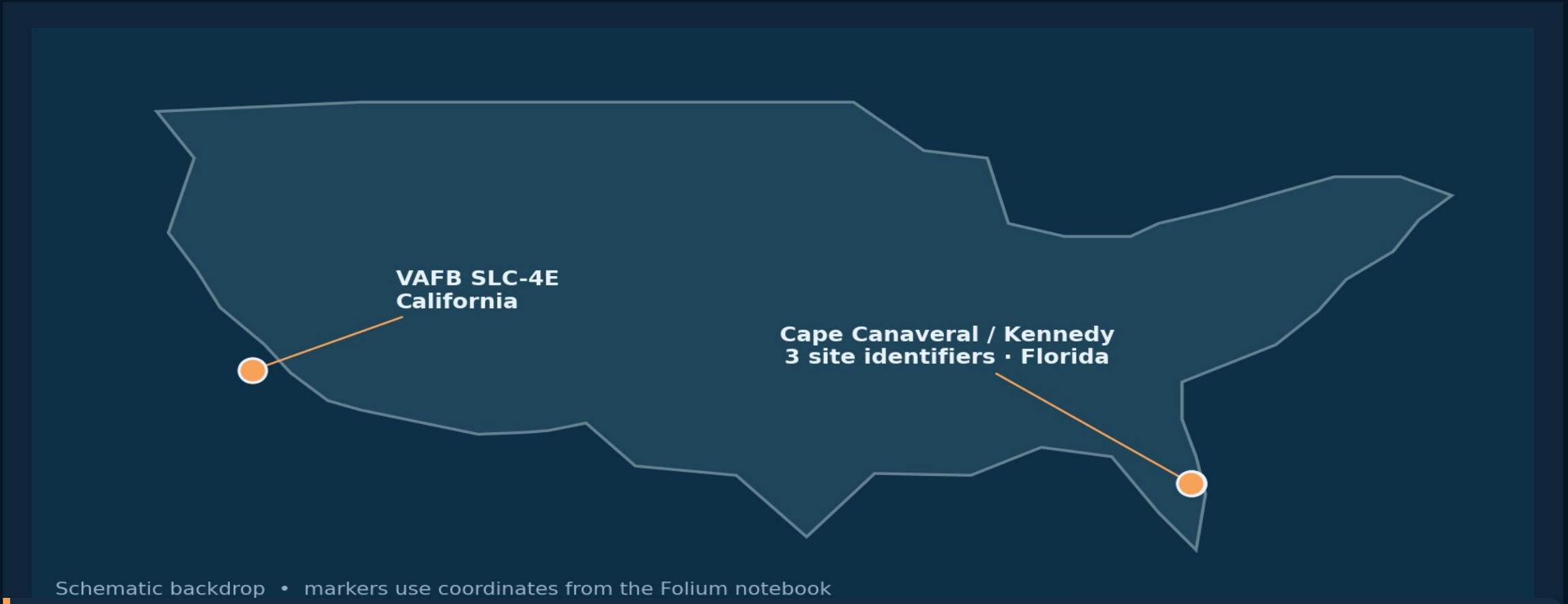
Launch-site distribution in the modelling sample



More observations come from Cape Canaveral than other locations; sample imbalance affects generalization.

Source: Data Wrangling(1).ipynb, LaunchSite.value_counts() (n=90).

Folium: geographic context of the launch sites



MarkerCluster groups overlapping launch records; green/red markers distinguish landing outcomes in the interactive notebook.

Source: Folium notebook, launch_sites_df, cells 18–46. Illustration uses coordinates recorded in the notebook.

Folium proximity analysis: what was measured?

4

Distinct launch-site IDs

1.01 km

Worked coastline example

Manually chosen map point

MousePosition

Map extension

Implementation

Circle + text labels for each site, marker clustering for launches, a cursor-coordinate plugin, PolyLine overlays, and haversine distances.

Interpretation boundary

Coastline/highway points were selected manually as approximate examples. These distances are illustrative, not verified nearest-neighbour distances.

Source: Folium notebook, cells 54–72.

The interactive dashboard: four completed tasks

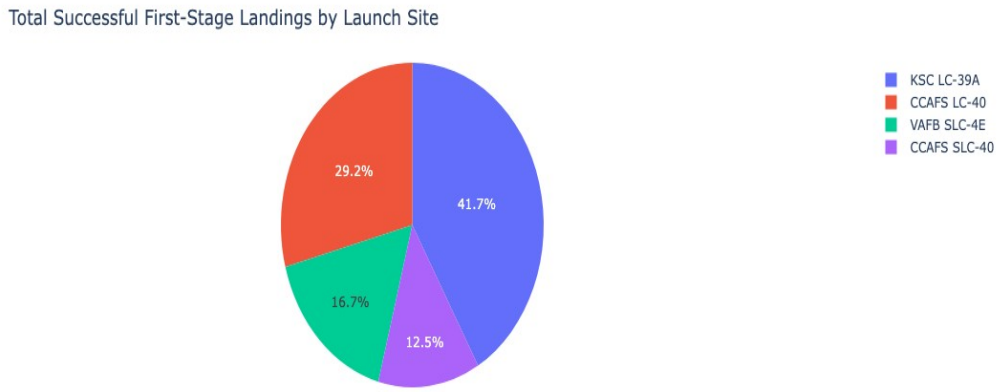
TASK 1	Launch-site dropdown: ALL + four locations
TASK 2	Pie chart callback: success volume / site outcomes
TASK 3	Payload mass range slider: 0–10,000 kg
TASK 4	Scatter callback: payload, class and booster version

Verified 25 OCT 2026: notebook server-start output and interactive dashboard visible in Safari at 127.0.0.1:8050.

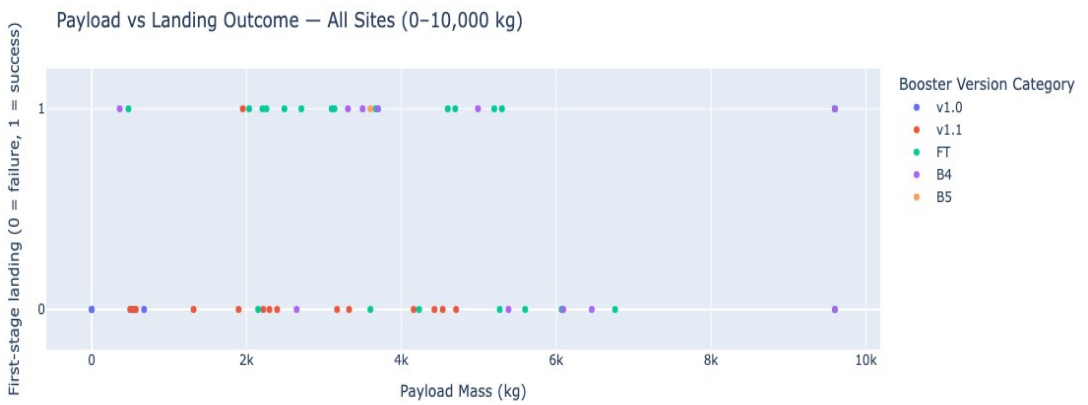
Sources: Dash notebook and IBM Dash laboratory PDF, pages 5–10.

Dashboard charts in the executed notebook

Launch site: ALL SITES **Payload slider: 0 — 10,000 kg**



Pie chart: successful first-stage landings by site

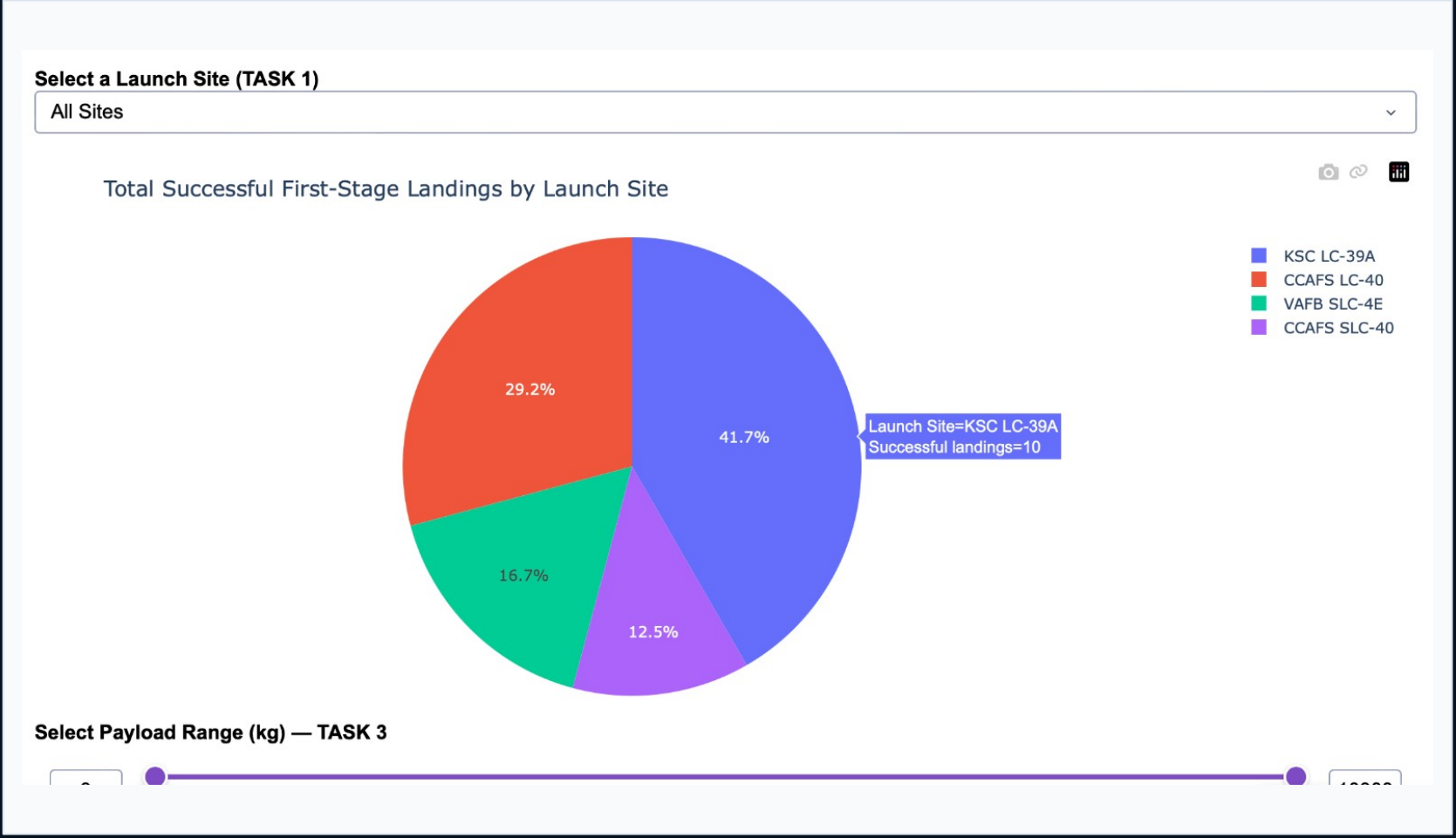


Scatter: payload, outcome and booster category

These are notebook-generated figures; the following two slides document the live Safari session.

Source: Plotly Dash notebook, saved outputs from callbacks for ALL sites and payload range.

Live dashboard: launch-site success overview



BROWSER EVIDENCE

Live application

25 October 2026 · Safari

127.0.0.1:8050

Local Dash server in notebook

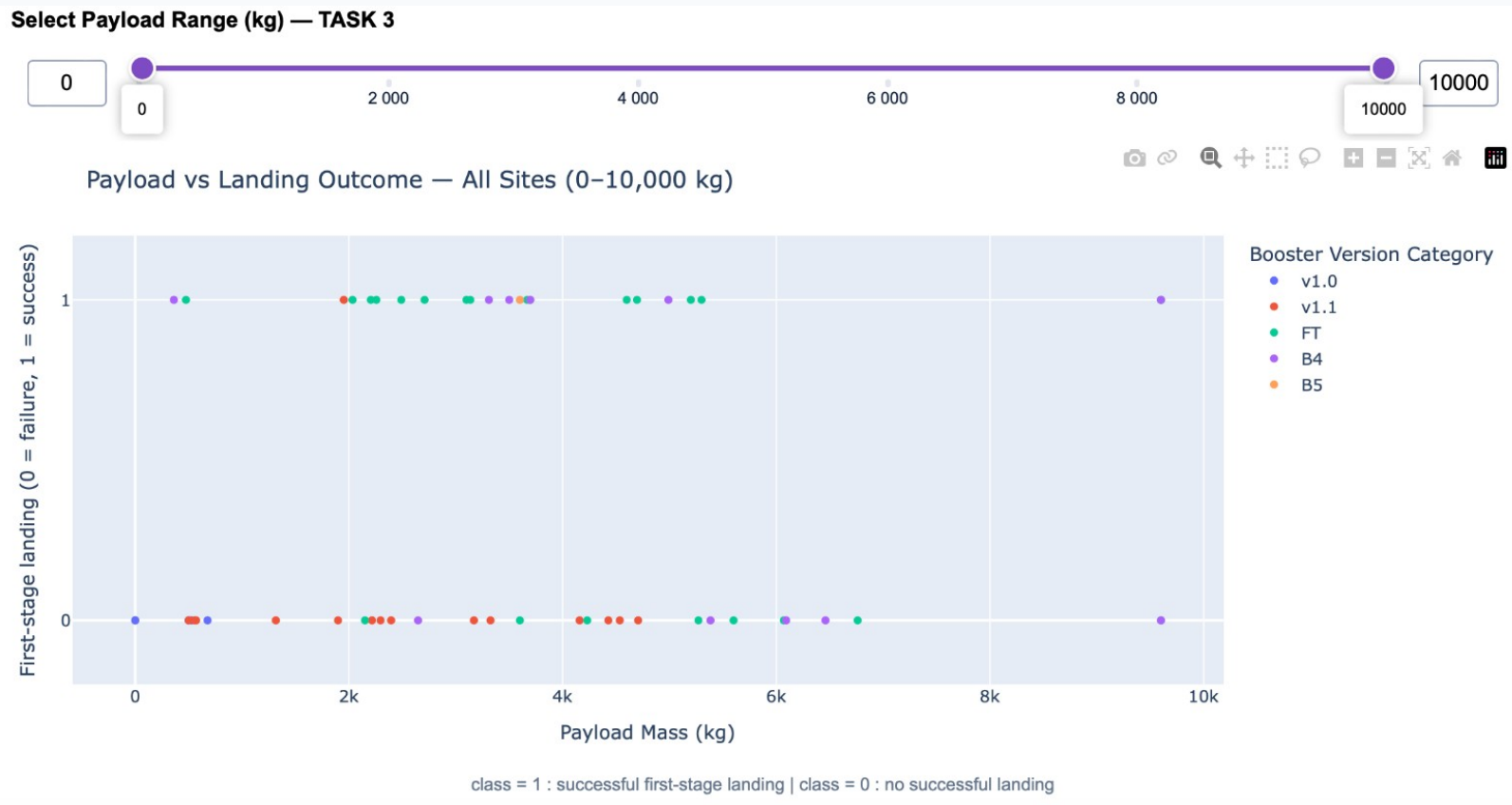
ALL SITES

Four launch sites available

KSC LC-39A: 10 successful landings (41.7% of successful landings in dashboard subset).

Source: supplied Safari screenshot at 15:21; notebook 7, executed server-start cell.

Live dashboard: payload and landing outcomes



BROWSER EVIDENCE

Interactive filters

25 October 2026 · Safari

0–10,000 kg

Interactive payload range

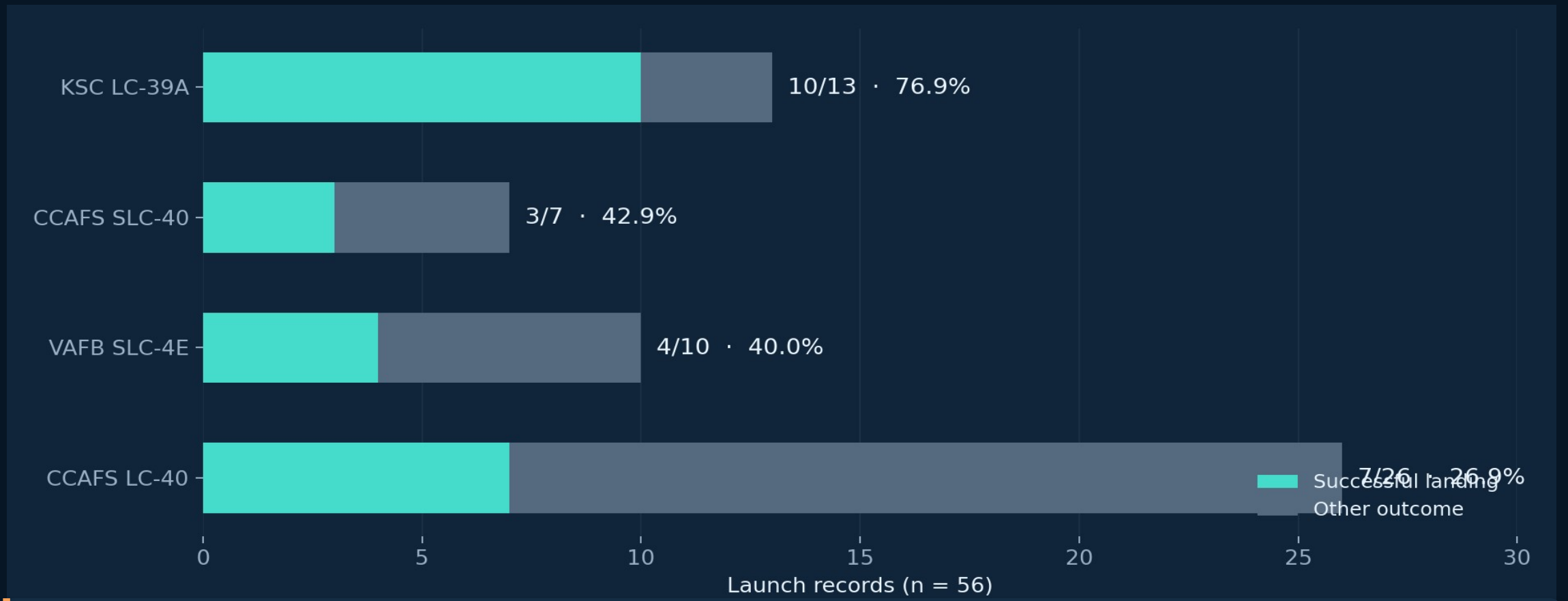
Class 0 / 1

Outcome by payload and booster

Visible booster categories: v1.0, v1.1, FT, B4 and B5. Values are historical.

Source: supplied Safari screenshot at 15:21; notebook 7, verified Dash callbacks.

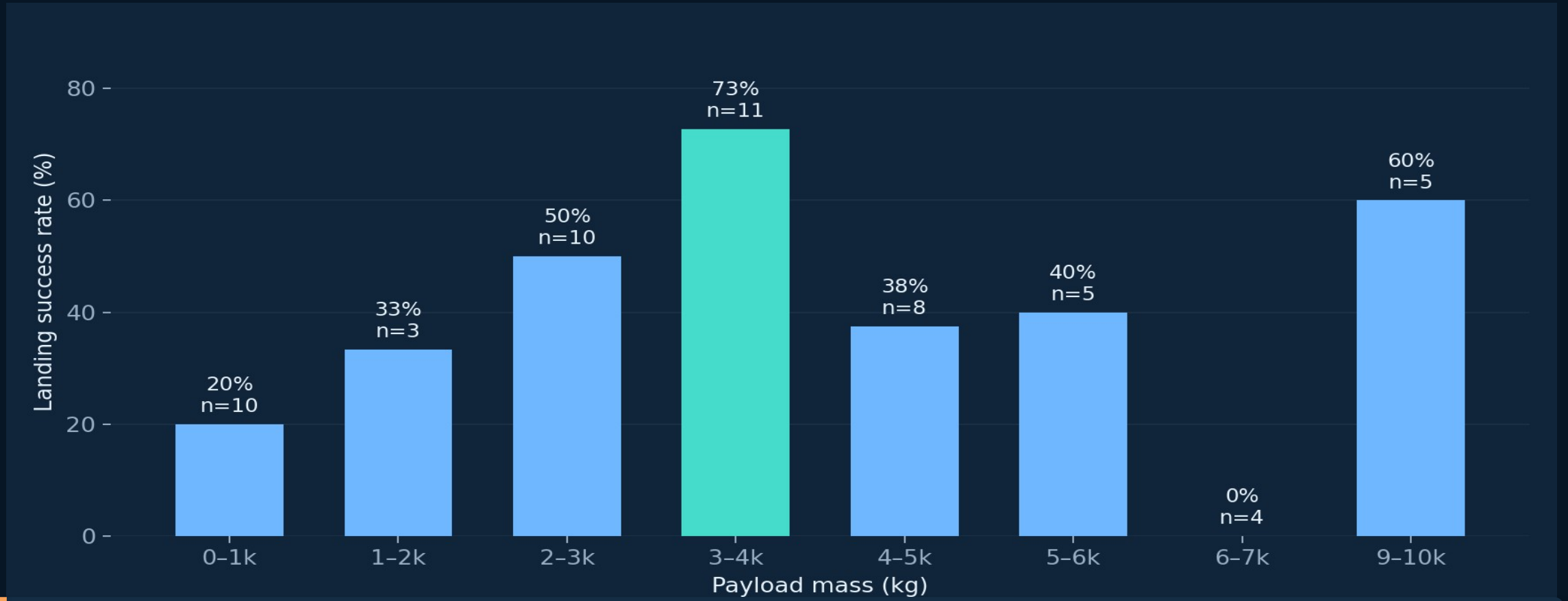
Which launch site performs best?



KSC LC-39A: 10 successful first-stage landings out of 13 (76.92%) — most successes and highest site rate.

Source: Dashboard notebook, site_stats output (n=56).

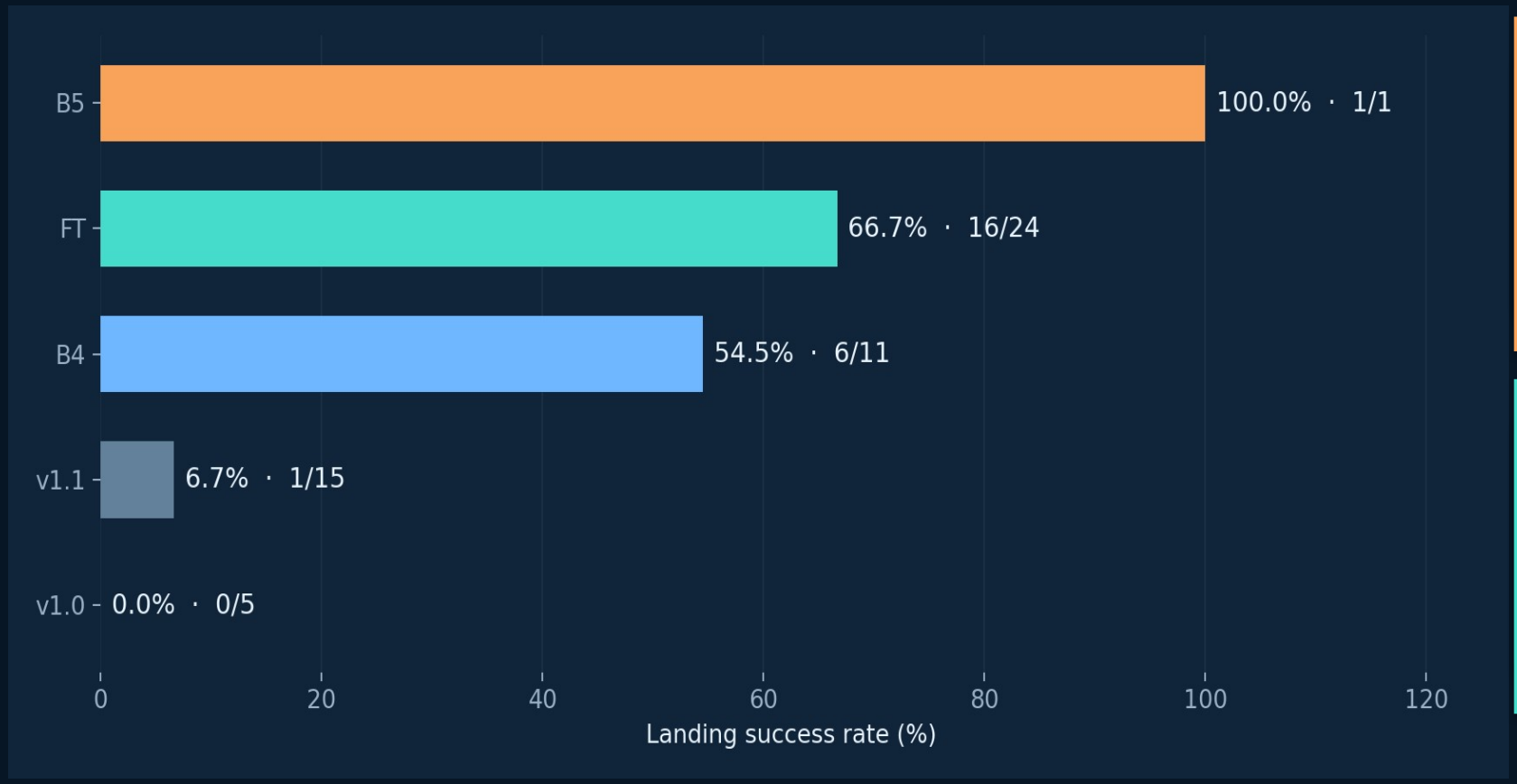
Does payload mass correlate with landing success?



Highest observed: 3,000–4,000 kg (8/11 = 72.73%); lowest: 6,000–7,000 kg (0/4 = 0%).

Source: Dashboard notebook, payload_stats output (1,000 kg bins, n=56).

Booster generations: rates need sample sizes



Top raw result

B5 shows 100%, but only one launch in this dataset (1/1). This is not evidence of a stable 100% population rate.

More informative comparison

FT records 16 successful landings in 24 launches (66.67%) — a materially larger sample.

Source: Dashboard notebook, booster_stats output (n=56).

Predictive workflow and model selection

01 FEATURES

83 columns in the exported modelling feature table. Numeric and one-hot encoded inputs.

02 SPLIT

Train/test split: 80/20 with random_state=2. Test holdout: 18 records.

03 SEARCH

10-fold GridSearchCV for Logistic Regression, SVM, Decision Tree and KNN.

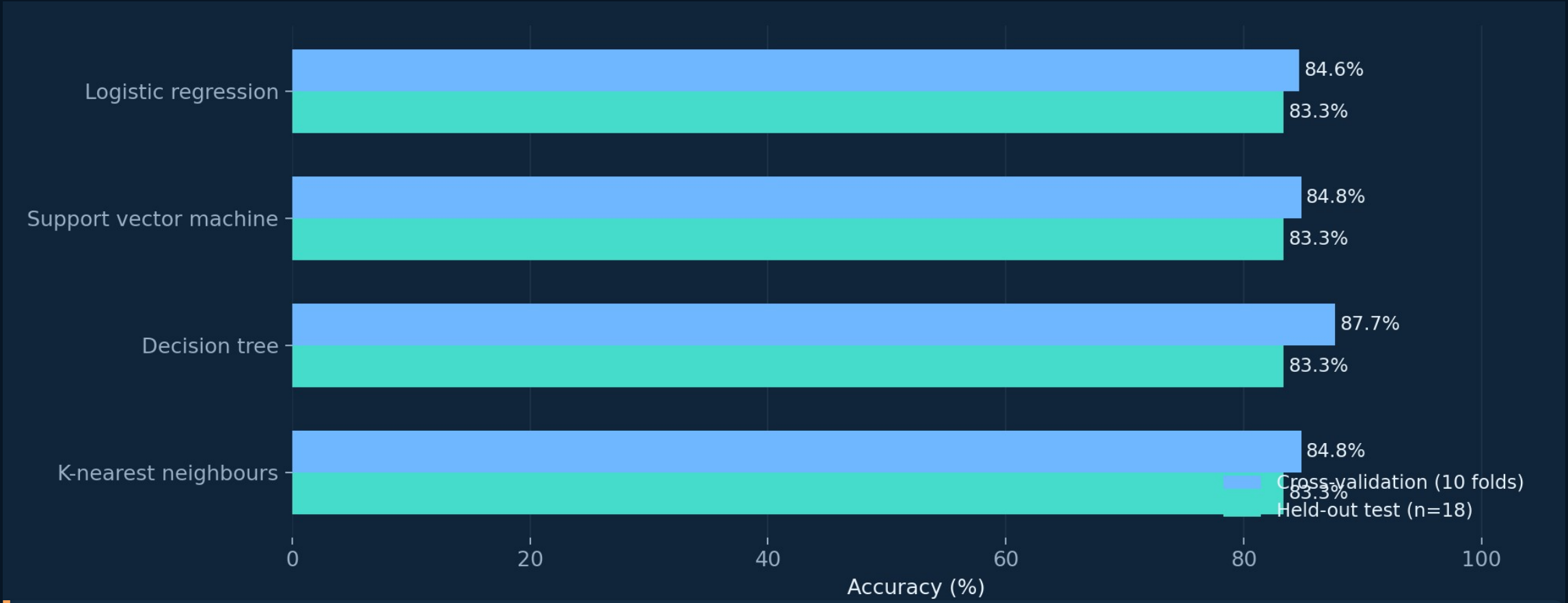
04 EVALUATE

Compare cross-validation scores with the held-out test accuracy and confusion matrix.

Methodological caution: the uploaded notebook fits StandardScaler before splitting; a pipeline would reduce leakage risk.

Source: Machine Learning Prediction notebook, cells 19–79.

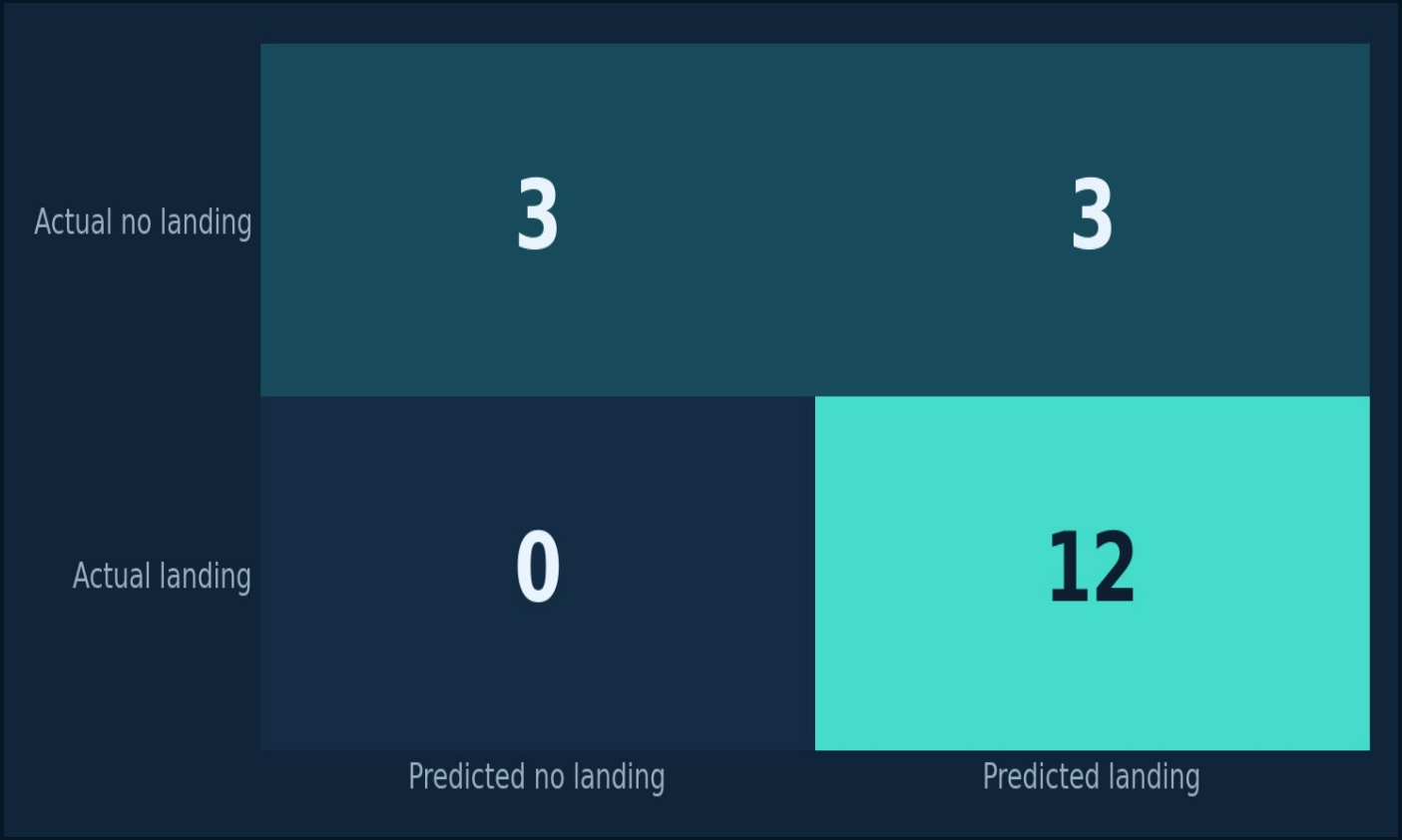
All four models tie on the independent test



15/18 correct for each model, equivalent to 83.33% test accuracy. CV leader: Decision Tree (87.68%).

Source: Machine Learning Prediction notebook, recorded cross-validation and held-out scores.

What the confusion matrices reveal



Predictions across 18 test flights

True positives: 12 | True negatives: 3
False positives: 3 | False negatives: 0

How to interpret

Accuracy is 83.33%, but the model incorrectly predicts landing in 3 of the 6 actual unsuccessful cases; specificity is 50%.

All four saved confusion-matrix plots show the same test classification counts.

Source: Machine Learning Prediction notebook, confusion-matrix images (all four models).

Conclusion: evidence, limits and next steps

Evidence

Historical landing results differ substantially by site and booster family. KSC LC-39A leads the dashboard subset.

Predictive result

Four algorithm families reach 83.33% on 18 holdout flights, with three false-positive landing calls each.

Limitations

Small and inconsistent cohorts; B5 has only one observation in Dash; scaling precedes split in the ML notebook.

Recommended improvements

Rebuild reproducible joined datasets, fit preprocessing within CV pipelines, test on newer out-of-time launches and report uncertainty.

Takeaway: a credible historical case study, not a production-grade predictive system.

Synthesis of submitted IBM SpaceX notebooks; historical datasets only.

Submission checklist and source references

Final presentation (PDF + PPTX)

This deck contains the business question, data lineage, acquisition, cleaning, SQL, EDA, Folium, Dash, classification and conclusions.

Project materials published on GitHub

The public repository contains eight course notebooks, a Python Dash app, results, evidence screenshots, and the final PDF and PowerPoint presentations.

GitHub repository URL

https://github.com/AkmMalek/IBM-SpaceX-Falcon9-Capstone

Live Dash runtime confirmed by saved notebook output and Safari screenshots. API execution remains unverified.

IBM Capstone rubric: PDF deck + GitHub repository with completed notebooks/Python files.

Technical appendix: provenance and methods

Web scraping	**121 × 11**	BeautifulSoup / historical launch page
Data Wrangling + EDA	**90 observations**	Outcome-to-Class mapping · one-hot features
SQL EDA	**101 rows**	SQLite + SELECT / GROUP BY / filtering
Folium	**4 sites**	Maps / clusters / coordinates / illustrative distance
Plotly Dash	**56 observations**	Dropdown + two callbacks; live Safari session confirmed
Machine learning	**90 total / 18 test**	10-fold CV and test confusion matrices
API notebook	**unexecuted**	API functions present; results not verified

Evidence: executed notebooks, screenshots of Dash running locally, and reproduced figures.